

Arakawa–Schubert (1974): goal of the paper

1

Large-scale models predict temperature and humidity averaged over large areas. They do not resolve the cumulus clouds inside, yet the clouds heat, cool, dry and moisten the air.

Goal

Compute the collective effect of the clouds from the large-scale fields themselves, so that the large-scale equations close. This is *cumulus parameterization*.

How: three steps

Step 1 · §2–3 · slide 2

Budgets of the air between the clouds: the clouds act only through three profiles in z .



Step 2 · §4–5 · slide 3

Clouds as entraining plumes of types λ : the three profiles follow from one spectrum.



Step 3 · §6–7 · slides 4–5

Quasi-equilibrium: the large-scale build-up of buoyancy balances its use by the clouds.

Unknowns left

3 profiles
 $M_c, D, \hat{l}$

1 spectrum
 $\mathcal{M}_B(\lambda)$

none:
closed

In a model, each time step: $\bar{s}, \bar{q} \rightarrow \mathcal{M}_B(\lambda) \rightarrow M_c, D, \hat{l} \rightarrow \bar{s}, \bar{q}$ at the next step

Step 1: clouds enter through three profiles

2

Budgets of the air between clouds: the model knows every term except M_c , D and $\hat{l}$.

$$\rho \partial_t \bar{s} = D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + \text{LS}_s$$

$$\rho \partial_t \bar{q} = D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + \text{LS}_q$$

Symbols

$s = c_p T + gz$, q dry static energy; vapour mixing ratio

$\bar{s}$, $\bar{q}$ large-scale means, i.e. the environment

$M_c = \sum_i \rho \sigma_i w_i$ cloud mass flux; σ_i , w_i : area fraction, vertical velocity of cloud i

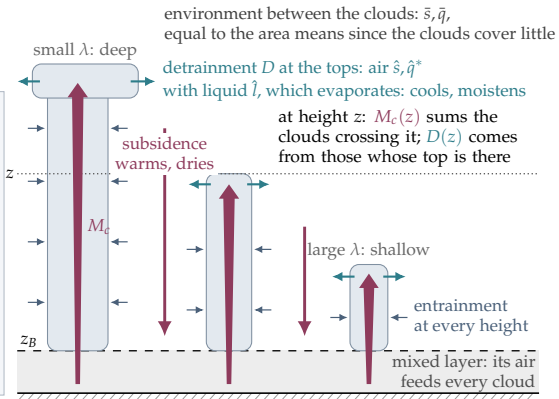
D mass detrained per unit volume and time

$\hat{s}$, $\hat{q}^*$ in the detrained air, saturated; set by $\bar{s}$, $\bar{q}$, $\hat{l}$

$\hat{l}$ liquid water in the detrained air

L , ρ latent heat of vaporization; air density

LS large-scale advection (+ radiation for s)



Unknowns left: three profiles, $M_c(z)$, $D(z)$ and $\hat{l}(z)$

Step 2: one spectrum gives all three profiles

3

Each cloud type λ mixes in outside air at every height, at its own rate. Given the environment, each type follows, and summing the types gives M_c , D and $\hat{l}$.

One cloud type λ

$$\partial_z \eta = \lambda \eta \Rightarrow \eta = e^{\lambda(z-z_B)}$$

$$\eta h_c = h_M + \lambda \int_{z_B}^z \eta \bar{h} dz'$$

All types together

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

$$D(z) = -\mathcal{M}_B(\lambda_D) \eta(z, \lambda_D) d\lambda_D / dz$$

$$\hat{l}(z) = l(z, \lambda_D)$$

Symbols

λ fractional entrainment rate, km^{-1} : the type label. Large λ : small, diluted cloud with a low top

$\eta(z, \lambda)$ mass flux of type λ , normalized to 1 at z_B

$h = s + Lq$ moist static energy: conserved also when vapour condenses

h_c, h_M h inside the cloud; in the subcloud mixed layer

z_B mixed-layer top: clouds start here with mixed-layer air

$z_D(\lambda)$ cloud top, where buoyancy vanishes: $h_c = \bar{h}^*$, the saturated h of the environment

$\lambda_D(z)$ type whose top is at height z

$l(z, \lambda)$ liquid water in type λ ; needs a rain model

$\mathcal{M}_B(\lambda)$ mass flux of type λ at z_B

Unknown left: one function, the spectrum $\mathcal{M}_B(\lambda)$

Step 3: quasi-equilibrium fixes the spectrum

4

No theory gives the number, lifetime and mass flux of the clouds of each type. The paper balances energy instead: the large scale builds up $A(\lambda)$, the clouds use it up.



$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \eta \underbrace{\frac{g(s_{vc} - \bar{s}_v)}{c_p \bar{T}}}_{\text{buoyancy}} dz$$

$$\frac{dA}{dt} = \int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) \simeq 0$$

$\simeq 0$: quasi-equilibrium, clouds adjust much faster than F changes.

Symbols

$A(\lambda)$ cloud work function: buoyant kinetic energy of type λ per unit mass flux

$s_v = s + c_p \bar{T}(0.608q - l)$, virtual static energy: vapour lightens the air, liquid water l weighs it

$s_{vc}, \bar{s}_v$ in the cloud; in the environment $g, \bar{T}$ gravity; environment temperature

$A > 0$ type λ can grow; $A(0) \approx \text{CAPE}$

$K(\lambda, \lambda')$ change of $A(\lambda)$ per unit $\mathcal{M}_B(\lambda')$; mostly negative

$F(\lambda)$ change of A by large-scale processes

No unknowns left: $\mathcal{M}_B(\lambda)$ solves an integral equation, and the system is closed

Step 3 tested: observations support quasi-equilibrium

5

Test with tropical data. If clouds did not respond, A would change as fast as the forcing (dashed line). The observed change of A stays near zero.

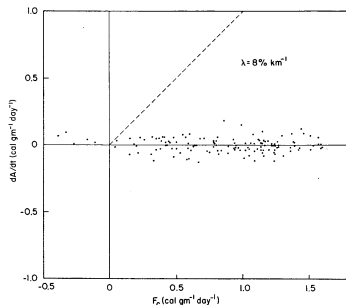


Fig. 13: observed dA/dt against F_C , for $\lambda = 8\% \text{ km}^{-1}$.

Two timescales

$$\tau_{\text{adj}} \sim 10^3 - 10^4 \text{ s}$$

15 min to 3 h

$$\tau_{\text{LS}} \gtrsim 10^5 \text{ s}$$

a day or more

Observed:

$$|dA/dt| \ll |F_C|$$

Symbols

F_C change of A that large-scale processes in the cloud layer would produce alone

dA/dt observed change of A

τ_{adj} time for the clouds to remove A

τ_{LS} time over which the large-scale forcing changes

cal g^{-1} energy per unit mass; 1

$\text{cal g}^{-1} \approx 4.2 \text{ kJ kg}^{-1}$

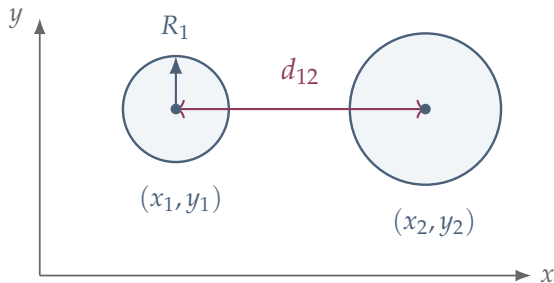
Clouds nearly cancel the forcing: A changes much less than F_C predicts

No 2D distribution of thermals

6

Our question: how are thermals arranged in a horizontal plane? The theory sorts clouds by type, not by position.

Spatial quantities (schematic)



Sought, not in the paper

R_i radius of thermal i at a fixed height

(x_i, y_i) position of its centre

d_{ij} distance between thermals i and j

Why the theory does not answer it

A spectrum over λ has no positions or distances.

$\mathcal{M}_B$ mixes cloud number, lifetime and single-cloud mass flux, never separated.

The environment is assumed horizontally uniform.

A mass-flux spectrum leaves the horizontal thermal field unspecified